

Introduction

In this practical session, you will set up the `classicmodels` sample database, analyse its schema using concepts from TP04 and TP05, and begin writing simple SQL statements to retrieve and manipulate data.

Learning Objectives

- Restore a MySQL database from a `.sql` file
- Identify entities, weak entities, relationships, and cardinality in a real schema
- Write basic `SELECT`, `WHERE`, `INSERT`, `UPDATE`, and `DELETE` statements

Submission Guidelines

- Write one answer for each exercise
- End each SQL statement with a semicolon
- Use clear indentation when needed

Part 1 – Setup & Restore

Step 1 – Install MySQL

1. Download and install **MySQL Community Server** (version 8.x).
2. Download and install **MySQL Workbench**.
3. Open MySQL Workbench and create a new connection:
 - Host: `localhost`
 - Port: `3306`
 - Username: `root`
 - Password: `password`
4. Click **Test Connection** to verify the setup.

Step 2 – Restore the `classicmodels` Database

The file `classicmodels.sql` is provided in the `SETUP` folder.

Option A – Command Line (recommended):

```
mysql -u root -p < classicmodels.sql
```

Option B – MySQL Workbench (query tab):

```
SOURCE /path/to/classicmodels.sql;
```

Option C – XAMPP / WAMP / LAMP:

```
mysql -u root -p classicmodels < classicmodels.sql
```

Step 3 – Verify the Restore

```
USE classicmodels;  
SHOW TABLES;
```

You should see these 8 tables:

Table	Key Columns
customers	customerNumber, customerName, city, country, creditLimit, salesRepEmployeeNumber
employees	employeeNumber, firstName, lastName, jobTitle, officeCode, reportsTo
offices	officeCode, city, country
orders	orderNumber, orderDate, status, customerNumber
orderdetails	orderNumber, productCode, quantityOrdered, priceEach
payments	customerNumber, checkNumber, paymentDate, amount
productlines	productLine, textDescription
products	productCode, productName, productLine, quantityInStock, buyPrice

Step 4 – Backup (for reference)

```
mysqldump -u root -p classicmodels > classicmodels_backup.sql
```

Part 2 – Schema Analysis (Guess the Design)

Using what you learned in TP04 and TP05, analyse the `classicmodels` schema above and answer the questions below. **Do not write SQL – write your answers in plain text or table form.**

Exercise 1 – Identify Entities

Look at the 8 tables. Decide which tables represent **strong entities** (they can exist on their own) and which are **weak entities** or **associative tables** (they depend on another table to exist).

Fill in the table below:

Table	Type	Reason
customers		
employees		
offices		
orders		
orderdetails		
payments		
productlines		
products		

Exercise 2 – Identify Relationships

List the relationships between the tables. For each relationship, write the two tables involved and describe what connects them.

Table A	Table B	Foreign Key / Connection
customers	employees	salesRepEmployeeNumber

One example has been filled in for you. Find and complete the remaining relationships.

Exercise 3 – Determine Cardinality

For each relationship you found in Exercise 2, determine the cardinality. Use the notation: 1:1, 1:N, or M:N.

Table A	Table B	Cardinality	Explanation
customers	employees	1:N	One employee handles many customers
employees	offices		
employees	employees		
orders	customers		
orderdetails	orders		
orderdetails	products		
payments	customers		
products	productlines		

Exercise 4 – Weak Entity Discussion

Answer these short questions in one or two sentences each:

1. Why can `orderdetails` be considered a weak entity?
2. What makes `productlines` a strong entity?
3. The `employees` table has a column `reportsTo` that references another row in the same table. What kind of relationship is this called?

Part 3 – Simple SELECT Queries

Always start each session with:

```
USE classicmodels;
```

Exercise 5 – Explore the Tables

Write SQL statements to show all rows and all columns from:

1. The `customers` table
2. The `products` table
3. The `employees` table

Exercise 6 – Select Specific Columns

Write SQL queries to:

1. Show only `customerName` and `city` from `customers`
2. Show only `productName` and `buyPrice` from `products`
3. Show only `firstName`, `lastName`, and `jobTitle` from `employees`

Exercise 7 – Filter with WHERE

Write SQL queries to:

1. Show all customers located in 'France'
2. Show all customers located in the 'USA'
3. Show all products with a `buyPrice` less than 50
4. Show all products with a `quantityInStock` greater than 5000
5. Show all employees whose `jobTitle` is 'Sales Rep'

Exercise 8 – Combined Conditions

Write SQL queries to:

1. Show customers in 'USA' with a `creditLimit` greater than 100000
2. Show products in the 'Classic Cars' product line with `buyPrice` greater than 80

Part 4 – INSERT, UPDATE, DELETE

Exercise 9 – Insert a New Customer

Insert the following row into the `customers` table:

Column	Value
<code>customerNumber</code>	999
<code>customerName</code>	My Test Company
<code>contactLastName</code>	Smith
<code>contactFirstName</code>	John
<code>phone</code>	0123456789
<code>addressLine1</code>	123 Test Street
<code>city</code>	Phnom Penh
<code>country</code>	Cambodia
<code>creditLimit</code>	50000

After inserting, write a `SELECT` to confirm the row was added.

Exercise 10 – Update a Row

Write SQL statements to:

1. Change the `city` of customer number 999 to 'Siem Reap'
2. Change the `creditLimit` of customer number 999 to 75000

Exercise 11 – Delete a Row

Write an SQL statement to delete the customer whose `customerNumber` is 999.

After deleting, write a `SELECT` to confirm the row no longer exists.

Expected Outcome

After completing this TP, students should be able to:

- restore a MySQL database from a `.sql` file
- identify entities, weak entities, relationships, and cardinality in a real schema
- write `SELECT` queries with and without `WHERE`
- insert, update, and delete rows in a real database